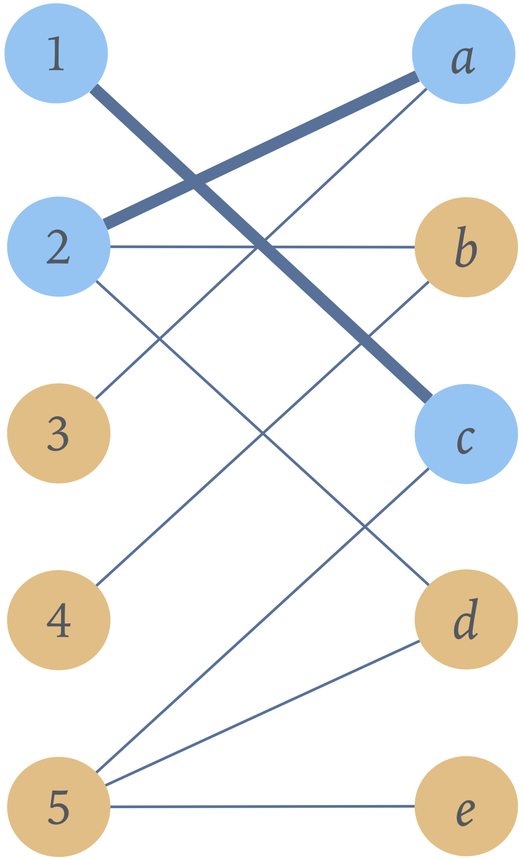
In-class Assignment L6-1 (Group of 3 or less): Hopcroft-Karp Algorithm

(25 minutes) Given a current matching $M = \{(1,c),(2,a)\}$ on the given graph G, work out Hopcroft-Karp algorithm to determine the maximum matching in G.

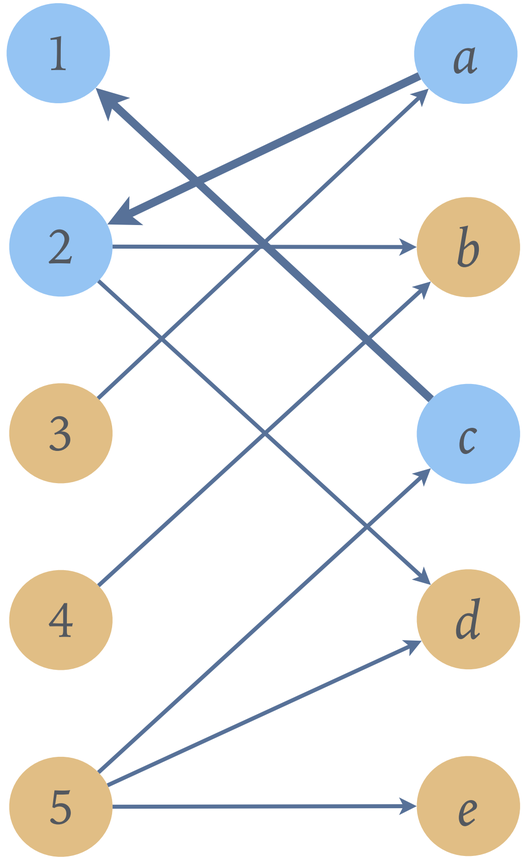
In each iteration of lines 3 and 4,

- Draw the digraph G_M
- Draw in DAG H obtained from running BFS on G_M
- Identify a maximal set of vertex-disjoint shortest M-augmenting paths from DAG H
- Show how the current matching M is improved in G.

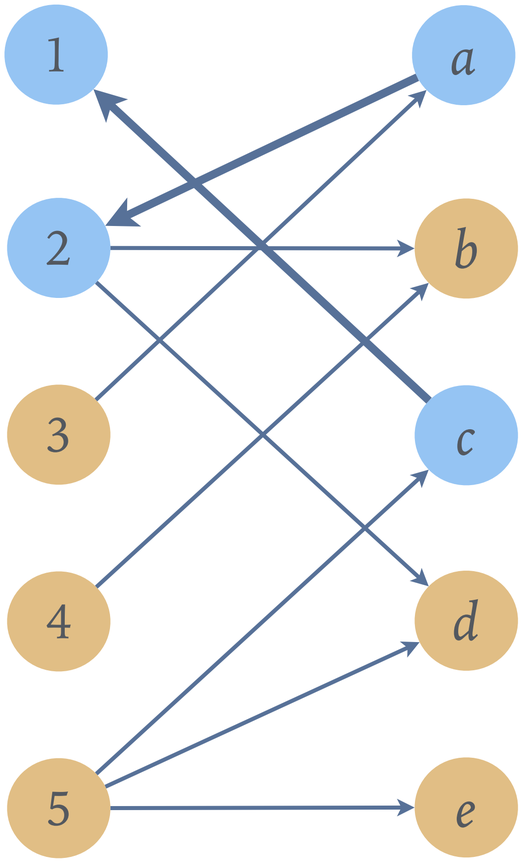
1st iteration



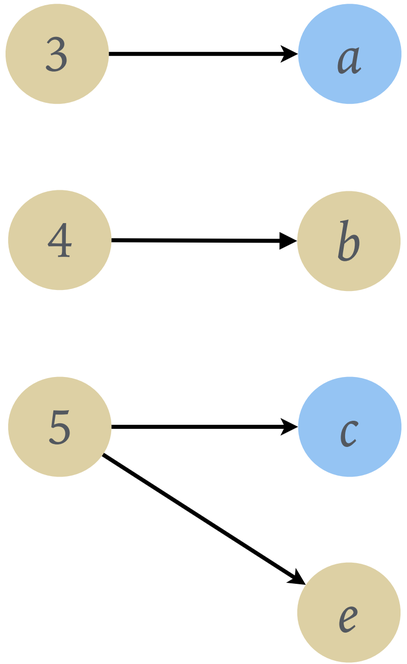
Digraph G_M



Solution

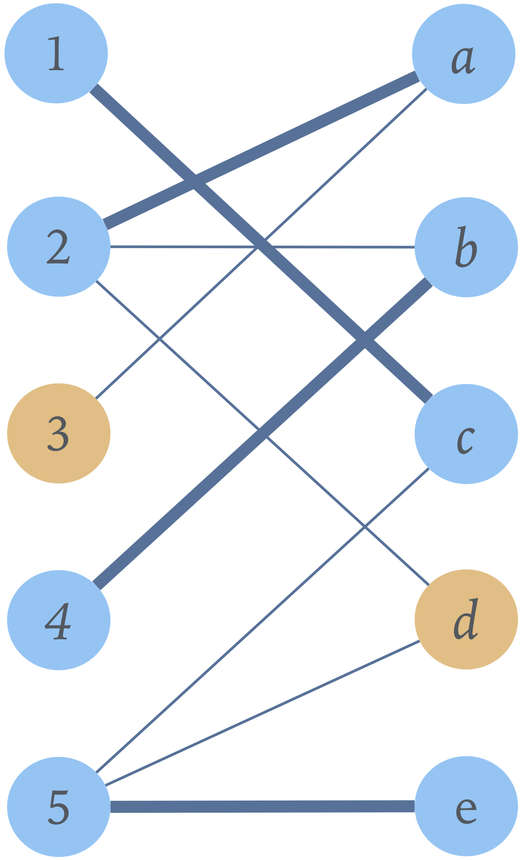


DAG $H = (V_H, E_H)$

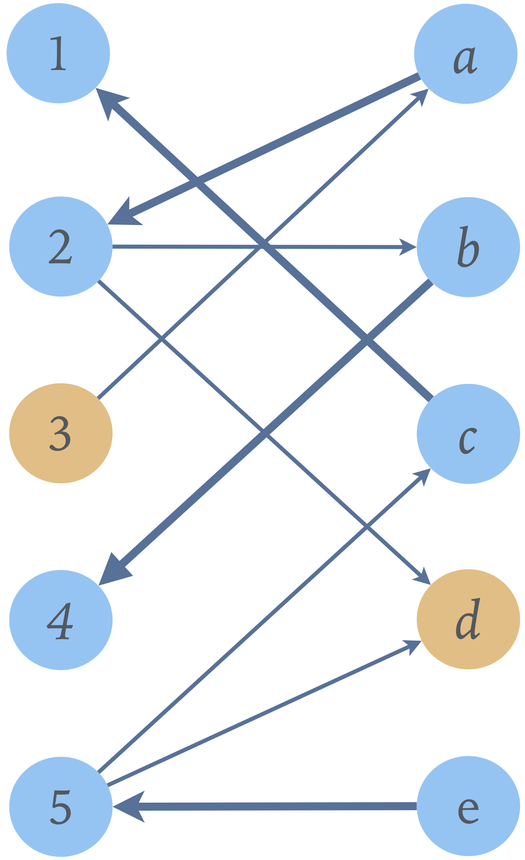


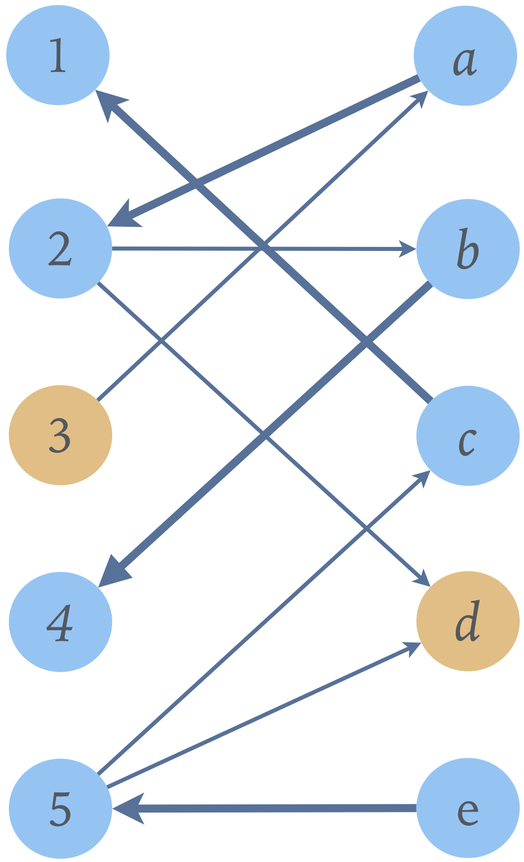
Maximal set of vertex-disjoint
shortest M-augmenting paths: P1 = {(4,b)}, P2={(5,e)}

2nd iteration

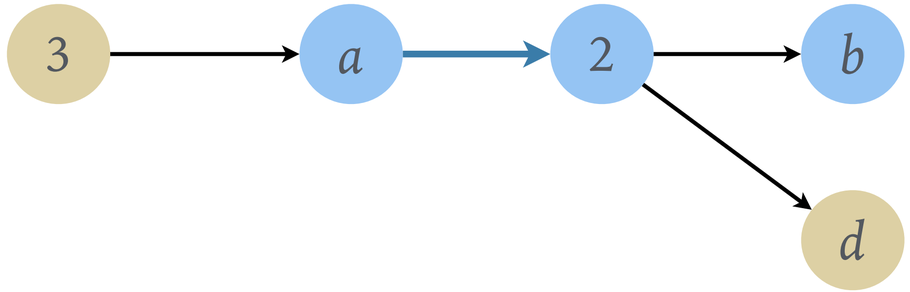


Digraph G_M





DAG $H = (V_H, E_H)$



Maximal set of vertex-disjoint
shortest M-augmenting paths: P1={(3,a), (5,e)}

3rd iteration

